

21/123 mRNA cis splicing, via spliceosome

59/498 mRNA processing

11/24 protein trimerization

14/131 regulation of peptidase

40/422 regulation of proteolysis

9/23 proteasomal ubiquitin-independent protein catabolic process

7/62 positive regulation of NF-kappaB transcription factor

8/49 steroid hormone receptor signaling pathway

61/607 RNA localization

24/140 obsolete chaperone-mediated protein folding

42/305 protein folding

11/43 ER-nucleus signaling pathway

9/45 obsolete regulation of telomerase

100/599 obsolete organic cyclic compound catabolic process

17/146 nucleoside monophosphate biosynthetic process

18/140 ribosomal small subunit biogenesis

58/591 obsolete ncRNA metabolic process

59/454 ribosome biogenesis

4/36 ribosomal small subunit assembly

20/117 ribosome assembly

20/125 ribosomal large subunit biogenesis

80/623 ribonucleoprotein complex biogenesis

3/10 translation reinitiation

86/275 cytoplasmic translation

111/684 amide biosynthetic process

7/21 formation of translation preinitiation complex

68/271 translational initiation

67/417 localization within membrane

56/247 establishment of protein localization to membrane

77/490 establishment of protein localization to organelle

3/29 response to interleukin-4

13/90 muscle system process

84/790 system process

16/112 circulatory system process

10/64 regulation of tube size

13/57 regulation of striated muscle contraction

18/105 regulation of muscle system process

29/217 regulation of system process

33/256 regulation of monoatomic ion transport

2/18 obsolete regulation of anion channel

11/102 positive regulation of transmembrane transport

75/630 monoatomic ion transmembrane transport

14/120 potassium ion transport

19/108 sodium ion transport

14/96 action potential

17/143 calcium ion transport

6/38 digestion

17/105 response to alkaloid

18/117 response to purine-containing compound

7/12 cGMP metabolic process

8/27 cyclic nucleotide metabolic process

11/65 obsolete cyclic-nucleotide-mediated signaling

25/148 obsolete second-messenger-mediated signaling

3/18 obsolete cGMP-mediated signaling

0/28 ammonium transmembrane transport

16/162 xenobiotic transport

2/29 acid secretion

33/334 monoatomic anion transport

4/43 phospholipid transport

16/162 lipid localization

1/12 regulation of sterol transport

5/10 cholesterol efflux

11/52 organic hydroxy compound transport

3/85 lipid homeostasis

6/49 monoatomic anion homeostasis

74/650 protein phosphorylation

3/26 obsolete activation of MAPKK

7/74 stress-activated protein kinase signaling cascade

45/384 protein autophosphorylation

5/72 neuromuscular junction development

5/31 response to glucagon

17/181 locomotory behavior

10/73 culmination involved in sorocarp development

4/29 regulation of DNA endoreduplication

46/403 actin filament-based process

65/589 anatomical structure formation involved in morphogenesis

22/235 sporulation

16/172 cellular component assembly involved in morphogenesis

61/538 cellular anatomical entity morphogenesis

13/190 external encapsulating structure organization

60/612 meiotic cell cycle

21/212 epithelial cell differentiation

4/36 central nervous system neuron differentiation

4/22 fat cell differentiation

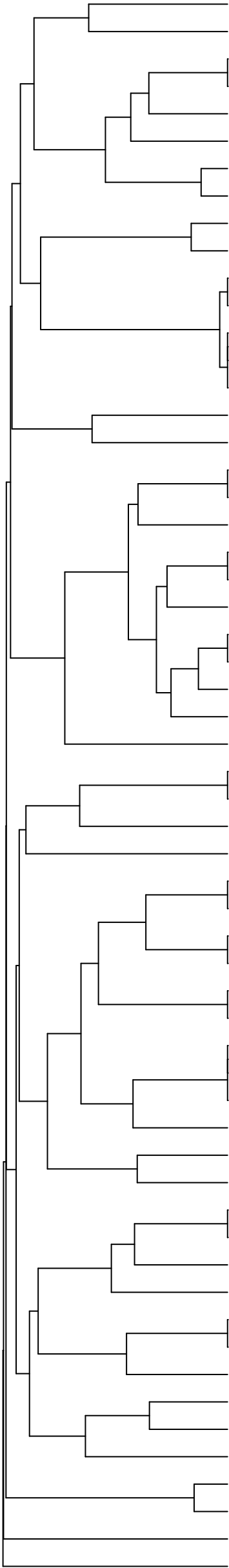
10/93 tissue homeostasis

p < 1e-04

p < 0.001

p < 0.01

SYMBIONT WINTER – Biological Process (BP)



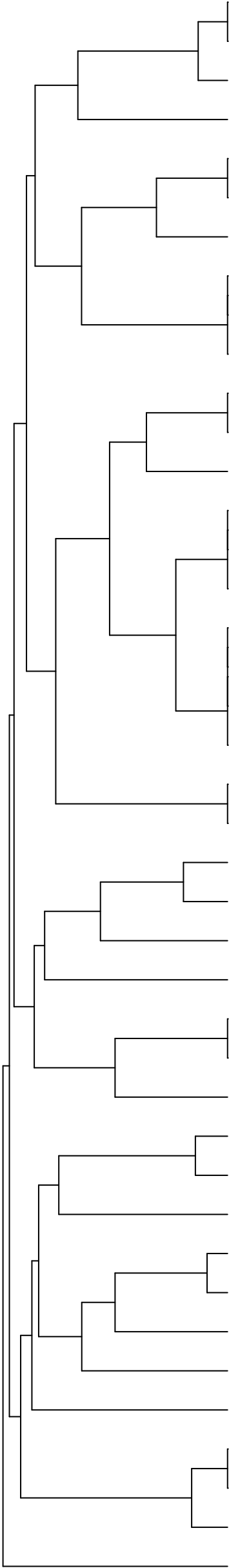
- 35/196 preribosome
- 4/39 sno(s)RNA-containing ribonucleoprotein complex
- 48/143 large ribosomal subunit
- 98/410 obsolete cytosolic part
- 57/205 obsolete polysome
- 23/84 small ribosomal subunit
- 16/60 rough endoplasmic reticulum membrane
- 17/86 rough endoplasmic reticulum
- 39/185 vesicle lumen
- 8/62 azurophil granule
- 87/775 intracellular protein-containing complex
- 14/49 cytosolic proteasome complex
- 10/33 proteasome storage granule
- 24/112 peptidase complex
- 6/14 proteasome core complex, alpha-subunit complex
- 5/25 endoplasmic reticulum-Golgi intermediate compartment membrane
- 15/96 endoplasmic reticulum lumen
- 60/565 ribonucleoprotein granule
- 30/258 nuclear ribonucleoprotein granule
- 8/53 U2-type catalytic step 2 spliceosome
- 16/113 Sm-like protein family complex
- 4/55 spliceosomal tri-snRNP complex
- 9/64 precatalytic spliceosome
- 4/31 U1 snRNP
- 33/271 spliceosomal complex
- 9/58 prespliceosome
- 10/48 U2 snRNP
- 6/24 catalytic step 1 spliceosome
- 2/31 extrinsic component of organelle membrane
- 9/128 extrinsic component of membrane
- 42/485 endosome
- 1/28 phragmoplast
- 4/22 neuron projection membrane
- 8/50 leading edge membrane
- 48/332 obsolete intrinsic component of plasma membrane
- 5/35 voltage-gated potassium channel complex
- 3/30 calcium channel complex
- 17/105 monoatomic ion channel complex
- 6/20 brush border membrane
- 15/103 cell projection membrane
- 38/341 plasma membrane region
- 27/189 apical part of cell
- 8/70 sarcolemma
- 15/90 I band
- 58/298 external encapsulating structure
- 2/22 spore wall
- 2/67 intracellular immature spore
- 3/23 amyloplast
- 8/99 cell cortex region
- 29/318 cell cortex
- 1/19 new growing cell tip
- 7/26 actin filament bundle
- 75/459 cell-cell junction
- 18/104 apoplast
- 21/126 mitochondrial protein-containing complex
- 8/27 obsolete respirasome
- 3/14 new mitotic spindle pole body
- 7/10 enzyme activator complex

p < 1e-04

p < 0.001

p < 0.01

SYMBIONT WINTER – Cellular Component (CC)



89/809 ATP-dependent

14/111 ATP-dependent, acting on DNA

13/230 nucleic acid conformation isomerase

19/142 polypeptide conformation or assembly isomerase

6/40 ABC-type transporter

46/337 active transmembrane transporter

3/19 xenobiotic transmembrane transporter

3/43 phospholipid transporter

7/67 lipid transporter

2/23 intramembrane lipid transporter

16/89 sodium ion transmembrane transporter

4/16 P-type sodium transporter

11/106 potassium ion transmembrane transporter

47/392 monoatomic cation transmembrane transporter

4/49 calcium channel

13/108 calcium ion transmembrane transporter

20/146 gated channel

29/216 channel

67/540 monoatomic ion transmembrane transporter

9/51 ligand-gated monoatomic ion channel

12/71 obsolete inorganic anion transmembrane transporter

22/216 monoatomic anion transmembrane transporter

10/46 cyclic nucleotide binding

77/648 adenylyl ribonucleotide binding

4/23 cyclic-nucleotide phosphodiesterase

16/110 molecular transducer

9/35 cyclic nucleotide-dependent protein kinase

65/627 phosphotransferase, alcohol group as acceptor

3/18 MAP kinase kinase kinase

85/574 protein-containing complex binding

3/14 protein kinase C inhibitor

2/22 alpha-1,4-glucan glucosyltransferase (ADP-glucose donor)

92/507 mRNA binding

4/11 opsonin binding

27/230 ubiquitin-like protein ligase binding

32/248 calcium ion binding

12/57 translation initiation factor

11/62 rRNA binding

2/22 5S rRNA binding

78/339 structural molecule

4/14 phosphotransferase, carboxyl group as acceptor

p < 1e-04

p < 0.001

p < 0.01

SYMBIONT WINTER – Molecular Function (MF)